

**THE MAHATMA GANDHI UNIVERSITY UNDER
GRADUATE VOCATIONAL PROGRAMMES
(HONOURS) SYLLABUS**

MGU-B.VOC. (Honours)

(2025 Admission Onwards)



Faculty: Science

**Expert Committee: Service Excellence and
Hospitality**

Subject: Public Health

**Mahatma Gandhi University
Priyadarshini Hills
Kottayam – 686560, Kerala, India**

Syllabus Index

Name of the Minor: Public Health

Semester: 1

Course Code	Title of the Course	Type of the Course	Credit	Hours/ week	Hour Distribution /week		
					L	P	O
MG1MPCPHS100	Introduction to Public Health	MPC	4	5	3	2	0

L — Lecture, P — Practical/Practicum , O — On the Job Training

Semester: 2

Course Code	Title of the Course	Type of the Course	Credit	Hours/ week	Hour Distribution /week		
					L	P	O
MG2MPCPHS100	Demography & Biostatistics of Public health	MPC	4	4	4	0	0

L — Lecture, P — Practical/Practicum , O — On the Job Training

Syllabus

Semester: 3

Course Code	Title of the Course	Type of the Course	Credit	Hours/ week	Hour Distribution /week		
					L	P	O
MG3MPCPHS200	Environmental and Occupational Health	MPC	4	5	3	2	0

L — Lecture, P — Practical/Practicum , O — On the Job Training

Semester: 4

Course Code	Title of the Course	Type of the Course	Credit	Hours/ week	Hour Distribution /week		
					L	P	O
MG4MPCPHS200	Maternal, Child, Adolescent & Geriatric Health	MPC	4	5	3	2	0

L — Lecture, P — Practical/Practicum , O — On the Job Training

Semester: 5

Course Code	Title of the Course	Type of the Course	Credit	Hours/ week	Hour Distribution /week		
					L	P	O
MG5MPCPHS300	Health Systems, Policy and Ethics	MPC	4	4	4	0	0

L — Lecture, P — Practical/Practicum , O — On the Job Training

Semester: 6

Course Code	Title of the Course	Type of the Course	Credit	Hours/ week	Hour Distribution /week		
					L	P	O
MG6MPCPHS300	Global Health and Public Health Informatics	MPC	4	4	4	0	0

L — Lecture, P — Practical/Practicum , O — On the Job Training



Mahatma Gandhi University

Kottayam

Faculty/ Discipline	Science				
Programme	Public Health				
Course Name	Introduction to Public Health				
Type of Course	MPC				
Course Code	MG1MPCPHS100				
Course Level	100				
Course Summary	This course provides a foundational understanding of health and disease, public health systems, promotion and prevention strategies, and India's public health landscape through integrated theoretical concepts and practical experiences.				
Semester	1	Credits		4	Total Hours
Course Details	Learning Approach	Lecture	Practical	OJT	
		3	1	0	75
Pre-requisites, if any	Nil				

COURSE OUTCOMES (CO)

CO NO.	EXPECTED COURSE OUTCOME	LEARNING DOMAINS*	PO NO.
CO1	Describe the fundamental concepts of public health, its scope, evolution, and the key determinants influencing health and disease.	U	1, 2,3,4,6,7,8,10
CO2	Explain the principles of disease prevention and health promotion and apply them to common public health conditions.	U, A	1, 2,3,4,6,7,8,10
CO3	Analyze the functions and approaches of public health systems and interpret strategies for disease control and inter-sectoral coordination.	An	1, 2,3, 4,5,6,7,8,9,10
CO4	Evaluate India's public health achievements, policies, and community-based programs, and create educational material for health promotion.	C	1, 2,3, 4,5,6,7,8,9,10
<i>Remember (K), Understand (U), Apply (A), Analyse (An), Evaluate (E), Create (C), Skill (S). Interest (I) and Appreciation (Ap).</i>			

MGU-B.VOC. (HONOURS)

CO-PO ARTICULATION MATRIX

Syllabus

CO/PO	PO 1	PO 2	PO 3	PO 4	PO 5	PO 6	PO 7	PO 8	PO 9	PO 10
CO 1	2	2	1	1	0	2	1	1	0	2
CO 2	1	2	1	1	0	2	1	1	0	3
CO 3	2	2	1	2	1	2	1	1	2	2
CO 4	1	2	2	3	1	3	2	2	1	2

'0' is No Correlation, '1' is Slight Correlation (Low level), '2' is Moderate Correlation (Medium level) and '3' is Substantial Correlation (High level).

COURSE CONTENT

Content for Classroom transaction (Units)

MODULE	UNITS	COURSE DESCRIPTION	Hrs.	CO. NO.
1	Foundations of Public Health		19	
	1.1	Definition, Scope, and Evolution of Public Health	4	CO1
	1.2	Concepts of Health and Disease	4	CO1
	1.3	Determinants of Health: Socioeconomic, Environmental, Biological	4	CO1
	1.4	Practical: Field visit to local PHC to understand primary healthcare delivery	7	CO1
2	Health Promotion and Disease Prevention		19	
	2.1	Levels of Prevention and Health Promotion	3	CO2
	2.2	Health Indicators and Health Status Measurement	4	CO2
	2.3	Natural History of Disease and Iceberg Phenomenon	4	CO2
	2.4	Practical: Group presentation on preventive strategies for common diseases	8	CO2
3	Public Health Practice and Systems		17	
	3.1	Core Functions and Essential Services of Public Health	4	CO3
	3.2	Public Health Approaches to Disease Control	4	CO3
	3.3	Inter-sectoral Coordination in Public Health	3	CO3
	3.4	Practical: Case study analysis of public health campaigns (e.g., polio eradication)	6	CO3
4	Epidemiology and Public Health in the Indian Context		20	
	4.1	Epidemiology: Definitions, aims, and applications in public health.	5	CO4
	4.2	Major Public Health Achievements in India	3	CO4
	4.3	Community Participation in Health Programs	3	CO4
	4.4	Practical: Preparation of health education material	9	CO4
5	Teacher Specific Content			

Teaching and Learning Approach	Classroom Procedure (Mode of transaction) • Direct Instructions: • Lecture • E Learning Interactive Instructions • Group Assignment • Library Work and Discussion • Practical
Assessment Types	MODE OF ASSESSMENT
	A. Continuous Comprehensive Assessment (CCA) Theory-25 Marks • Internal Test • Assignment/Oral presentation • Quiz • In-class discussion and involvement Practical-15 Marks • Internal Test • Record • Survey report
	A. End Semester Evaluation (ESE) Theory -50 Marks Duration of Examination 1.30 hrs Pattern of examination for Theory Non-MCQ • Section A - Objective type - 10/10 (10x1=10 marks) • Section B - Short Answer - 10/12 (10x2=20 marks) • Section C - Short Essay - 5/7 (5x4=20 marks) Practical - 35Marks • Presentation -10 marks • Viva- 10 marks • Field visit report/Case study-10 marks • Record-5marks Duration of Examination - NA

REFERENCES

1. **Park, K. (2023).** *Park's Textbook of Preventive and Social Medicine* (27th ed.). Jabalpur: Banarsidas Bhanot.
2. **Schneider, M. J. (2021).** *Introduction to Public Health* (6th ed.). Burlington, MA: Jones & Bartlett Learning.
3. **Friis, R. H., & Sellers, T. A. (2020).** *Epidemiology for Public Health Practice* (6th ed.). Burlington, MA: Jones & Bartlett Learning.
4. **Ministry of Health and Family Welfare, Government of India. (2017).** *National Health Policy 2017*. New Delhi. Retrieved from <https://main.mohfw.gov.in/>
5. **Indian Council of Medical Research (ICMR). (2009).** *Ethical Guidelines for Biomedical Research on Human Participants*. New Delhi: ICMR.

SUGGESTED READINGS

1. **Merson, M. H., Black, R. E., & Mills, A. J. (2012).** *Global Health: Diseases, Programs, Systems, and Policies* (3rd ed.). Burlington, MA: Jones & Bartlett.
2. **Bhatia, R. (2020).** *Essentials of Public Health: A Practical Guide*. New Delhi: CBS Publishers.
3. **Gupta, M. C., & Mahajan, B. K. (2019).** *Textbook of Preventive and Social Medicine* (4th ed.). New Delhi: Jaypee Brothers.
4. **Sundaram, K. R., Dwivedi, S. N., & Sreenivas, V. (2010).** *Medical Statistics: Principles and Methods*. New Delhi: BI Publications.
5. **Detels, R., Gulliford, M., Karim, Q. A., & Tan, C. C. (Eds.). (2015).** *Oxford Textbook of Global Public Health* (6th ed.). Oxford University Press.

MGU-B.VOC. (HONOURS)

Syllabus



Mahatma Gandhi University

Kottayam

Faculty/ Discipline	Science				
Programme	Public Health				
Course Name	Demography and Biostatistics of Public Health				
Type of Course	MPC				
Course Code	MG2MPCPHS100				
Course Level	100				
Course Summary	This course foundational knowledge of population dynamics, demographic indicators, statistical methods, and probability distributions essential for analyzing and interpreting public health data.				
Semester	2	Credits		4	Total Hours
Course Details	Learning Approach	Lecture	Practical	OJT	
		4	0	0	
Pre- requisites, if any	Nil.				

COURSE OUTCOMES (CO)

CO No.	Expected Course Outcome	Learning Domains	PO No
1	Describe key demographic concepts, sources of population data, and interpret basic measures of population change.	U	1,2,3,4,6,7,10
2	Analyze population structure using demographic indicators such as fertility, mortality, and morbidity; construct population pyramids and life tables.	An ,U	1,2,3,4,6,7,8,10
3	Apply biostatistical methods for data classification, summarization, and graphical representation in public health contexts.	A , U	1,2,3,4,6,9,10
4	Interpret probability and statistical distributions and explain their application in health research and	U ,An	1,2,3,4,6,10
Remember (K), Understand (U), Apply (A), Analyse (An), Evaluate (E), Create (C), Skill (S), Interest (I) and Appreciation (Ap)			

MGU-B.VOC. (HONOURS)

CO-PO ARTICULATION MATRIX

CO/ PO	PO 1	PO 2	PO 3	PO 4	PO 5	PO 6	PO 7	PO 8	PO 9	PO 10
CO 1	1	2	1	1	0	2	1	0	0	2
CO 2	2	2	1	1	0	2	1	1	0	2
CO 3	1	3	1	2	0	2	0	0	1	3
CO 4	2	3	1	1	0	2	0	0	0	3

'0' is No Correlation, '1' is Slight Correlation (Low level), '2' is Moderate Correlation (Medium level) and '3' is Substantial Correlation (High level).

COURSE CONTENT

Content for Classroom transaction (Units)

Module	Units	Course description	Hrs	CO No.
1	Foundations of Demography		15	
	1.1	Introduction to Demography: Definitions	5	CO1
	1.2	Sources of Demographic Data: Census	5	CO1
	1.3	Measures of Population Change: Birth Rate	5	CO1
2	Demographic Analysis and Indicators		15	
	2.1	Population Structure and Composition	5	CO2
	2.2	Population Pyramid and Life Table Construction	5	CO2
	2.3	Fertility, Mortality, and Morbidity Indicators	5	CO2
3	Introduction to Biostatistics		15	
	3.1	Introduction to Biostatistics	5	CO3
	3.2	Types of Data and Data Collection	5	CO3
	3.3	Descriptive Statistics	5	CO3
4	Basics of Probability and Distributions		15	
	4.1	Concepts of probability and its rules	5	CO4
	4.2	Introduction to normal, binomial, and Poisson distributions	5	CO4
	4.3	Importance of probability distributions in health research	5	CO4
5	Teacher Specific Content			

Teaching and Learning Approach	Classroom Procedure (Mode of transaction) Direct instructions Lecture E-Learning Interactive instructions Group assignments Group discussion
Assessment Types	MODE OF ASSESSMENT
	A. Continuous Comprehensive Assessment (CCA) Theory : 30 marks • Internal test • Assignment/ oral presentation • Quiz • In class discussion and involvement
	B. End Semester Evaluation (ESE) Theory -70 Marks Duration of Examination: 2 hrs Pattern of examination for Theory :Non-MCQ • Section A - Short Answer – 10/12 (10x2=20 marks) • Section B - Short Essay - 6/9 (6x5=30 marks) • Section C - Essay – 2/4 (2x10=20 marks)

REFERENCES

1. **Park, K. (2023).** *Park's Textbook of Preventive and Social Medicine* (27th ed.). Jabalpur: Banarsidas Bhanot.
2. **Sundaram, K. R., Dwivedi, S. N., & Sreenivas, V. (2010).** *Medical Statistics: Principles and Methods*. New Delhi: BI Publications.
3. **Mahajan, B. K. (2020)** *Methods in Biostatistics for Medical Students and Research Workers* (9th ed.). New Delhi: Jaypee Brothers.
4. **Census of India. (Latest Edition).** *Primary Census Abstracts and Population Reports*. Government of India.
5. **World Health Organization (WHO).** (2006). *Core Health Indicators: Definitions and Meta data*.

SUGGESTED READINGS

1. **Daniel, W. W., & Cross, C. L. (2018).** *Biostatistics: A Foundation for Analysis in the Health Sciences* (11th ed.). Hoboken, NJ: Wiley.
2. **Gerstman, B. B. (2014).** *Basic Biostatistics: Statistics for Public Health Practice* (2nd ed.). Burlington, MA: Jones & Bartlett Learning.
3. **Gupta, M. C., & Mahajan, B. K. (2019).** *Textbook of Preventive and Social Medicine* (4th ed.). New Delhi: Jaypee Brothers.
4. **Pagano, M., & Gauvreau, K. (2018).** *Principles of Biostatistics* (2nd ed.). Boca Raton, FL: CRC Press.
5. **Siegel, J. S., & Swanson, D. A. (2004).** *The Methods and Materials of Demography* (2nd ed.). San Diego: Elsevier Academic Press.

MGU-B.VOC. (HONOURS)

Syllabus



Mahatma Gandhi University

Kottayam

Faculty/ Discipline	Science				
Programme	Public Health				
Course Name	Environmental and Occupational Health				
Type of Course	MPC				
Course Code	MG3MPCPHS200				
Course Level	200				
Course Summary	This course introduces students to the foundational concepts of environmental and occupational health, with a special focus on their implications for nutrition, public health, and safety.This course also covers occupational hazards, ergonomic principles, and workplace safety regulations.				
Semester	3	Credits		4	Total Hours
Course Details	Learning Approach	Lecture	Practical	OJT	
		3	1	0	
Pre-requisites, if any	Nil				

COURSE OUTCOMES (CO)

CO No.	Expected Course Outcome	Learning Domains	PO No
1	Explain the concepts, scope, and significance of environmental and occupational health, including water quality and air pollution control measures.	U	1,2,3,6,7,8,10
2	Identify sources, types, and impacts of pollution and waste, and interpret their implications on public health and food safety.	U	1,2,3,4,6,7,10
3	Assess occupational hazards, diseases, and workplace safety measures with emphasis on ergonomic practices and preventive strategies.	A, An	1,2,3,4,5,6,7,10
4	Discuss major environmental health legislations, climate change impacts, and disaster preparedness strategies in the context of public health response.	U, E	1,2,3,4,5,6,7,8,9,10
*Remember(K), Understand(U), Apply(A), Analyse(An), Evaluate(E), Create(C), Skill(S), Interest(I) and Appreciation (Ap)			

MGU-B.VOC. (HONOURS) CO-PO ARTICULATION MATRIX

CO/ PO	PO 1	PO 2	PO 3	PO 4	PO 5	PO 6	PO 7	PO 8	PO 9	PO 10
CO 1	1	3	1	0	0	3	1	1	0	3
CO 2	1	3	1	1	0	3	1	0	0	3
CO 3	3	3	2	2	1	3	1	0	0	3
CO 4	2	3	2	3	1	2	2	2	1	3

‘0’ is No Correlation, ‘1’ is Slight Correlation (Low level), ‘2’ is Moderate Correlation (Medium level) and ‘3’ is Substantial Correlation (High level).

COURSE CONTENT

Content for Classroom transaction (Units)

Module	Units	Course description	Hrs	CO No.
1	Introduction to Environmental Health		17	
	1.1	Definition and Scope of Environmental Health	3	CO1
	1.2	Environmental Sanitation and Water Quality Standards	3	CO1
	1.3	Air Pollution: Sources, Effects, and Control Measures	3	CO1
	1.4	Practical: Testing and analysis of drinking water samples	8	CO1
2	Waste, Pollution, and Food Safety		20	
	2.1	Solid Waste and Biomedical Waste Management	4	CO2
	2.2	Noise Pollution and Radiation Hazards	4	CO2
	2.3	Food Safety and Public Health Regulations	4	CO2
	2.4	Practical: Field visit to a waste management unit	8	CO2
3	Occupational Health and Workplace Safety		19	
	3.1	Occupational Health: Concepts and Occupational Hazards	4	CO3
	3.2	Occupational Diseases and Injuries	4	CO3
	3.3	Workplace Safety and Ergonomics	4	CO3
	3.4	Practical: Health survey of a small occupational setting	7	CO3
4	Policy, Climate, and Disaster Preparedness		19	

	4.1	Environmental Health Legislation in India	4	CO4
	4.2	Climate Change and Health Impacts	4	CO4
	4.3	Disaster Preparedness and Public Health Response	4	CO4
	4.4	Practical: Conduct a mini survey in the campus	7	CO4
5	Teacher Specific Content			

Teaching and Learning Approach	Classroom Procedure (Mode of transaction) Direct Instructions: ● Lecture ● E-learning Interactive Instructions: ● Group Assignment ● Library Work and Group Discussion ● Practical			
	MODE OF ASSESSMENT MGU-B.VOC. (HONOURS) B. Continuous Comprehensive Assessment (CCA) Theory-25 Marks ● Internal Test ● Assignment/Oral presentation ● Quiz ● In-class discussion and involvement Practical-15 Marks ● Internal Test ● Record ● Survey report			

	C. End Semester Evaluation (ESE) Theory -50 Marks Duration of Examination 1.30 hr Pattern of examination for Theory Non-MCQ Section A - Objective type - 10/10 (10x1=10 marks) Section B - Short Answer - 10/12 (10x2=20 marks) Section C - Short Essay – 5/7 (5x4=20 marks) Practical-35Marks • Presentation -10 marks • Viva- 10 marks • Field visit report/Case study-10 marks • Record-5marks Duration of Examination -NA
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REFERENCES

1. **Park, K.** (2023). *Preventive and Social Medicine* (27th ed.). M/s Banarsidas Bhanot Publishers. Covers environmental sanitation, air & water quality, occupational health, pollution, public health laws.
2. **Friis, R. H.** (2018). *Essentials of Environmental Health* (3rd ed.). Jones & Bartlett Learning. Excellent for understanding environmental exposures, water/air pollution, waste management.
3. **World Health Organization (WHO)** (2021). *Occupational Health: A Manual for Primary Health Care Workers*.
4. **Yassi, A., Kjellstrom, T., De Kok, T., & Guidotti, T.** (2001). *Basic Environmental Health*. Oxford University Press.

SUGGESTED READINGS

1. **Murray, R. K., et al.** (2018). *Harper's Illustrated Biochemistry* (31st ed.). McGraw-Hill. Useful for biochemical aspects related to pollution and detoxification.
2. **Swaminathan, M.** (2012). *Advanced Textbook on Food & Nutrition* (Vol. I & II) Bangalore Printing & Publishing Co. For food safety, toxicology, and nutrient interactions with environmental exposures.
3. **Sivashanmugam, K., & Indira, K.** (2009). *Introduction to Environmental Engineering and Science*. Pearson Education.



Mahatma Gandhi University

Kottayam

Faculty/ Discipline	Science				
Programme	Public Health				
Course Name	Maternal, Child, Adolescent, and Geriatric Health				
Type of Course	MPC				
Course Code	MG4MPCPHS200				
Course Level	200				
Course Summary	This course provides a comprehensive overview of health concerns and services across critical stages of the human life span - from pregnancy through old age.				
Semester	4	Credits		4	Total Hours
Course Details	Learning Approach	Lecture	Practical	OJT	
		3	1	0	75
Pre-requisites, if any	Nil				

COURSE OUTCOMES (CO)

CO No.	Expected Course Outcome	Learning Domains	PO No
1	Explain the components of maternal health services including antenatal, intranatal, postnatal care, and management of high-risk pregnancies.	U,A	1,2,3,4,5,6,7,10
2	Assess child health through growth monitoring, immunization tracking, and early identification of common childhood illnesses.	A,An	1,2,4,5,6,7,10
3	Interpret major adolescent health concerns and design awareness or educational interventions for adolescent wellbeing.	U,A	1,2,3,4,5,6,10
4	Describe ageing-related health challenges and available geriatric services, and evaluate psychosocial needs of elderly populations.	U,E	1,2,4,6,7,8,9,10
*Remember(K), Understand(U), Apply(A), Analyse(An), Evaluate(E), Create(C), Skill(S), Interest(I) and Appreciation (Ap)			

CO-PO ARTICULATION MATRIX

CO/PO	PO 1	PO 2	PO 3	PO 4	PO 5	PO 6	PO 7	PO 8	PO 9	PO 10
CO 1	2	3	1	1	1	1	1	0	0	3
CO 2	1	3	0	1	1	3	1	0	0	3
CO 3	1	3	1	3	2	2	0	0	0	3
CO 4	1	3	0	2	0	3	2	2	1	3

‘0’ is No Correlation, ‘1’ is Slight Correlation (Low level), ‘2’ is Moderate Correlation (Medium level) and ‘3’ is Substantial Correlation (High level).

COURSE CONTENT

Content for Classroom transaction (Units)

MOD ULE	UNITS	COURSE DESCRIPTION	Hrs.	CO. NO.
1	Maternal Health		17	
	1.1	Introduction to Reproductive and Child Health (RCH)	3	CO1
	1.2	Antenatal, Intranatal, and Postnatal Care	4	CO1
	1.3	High-risk Pregnancies and Maternal Mortality	5	CO1
	1.4	Practical: Antenatal check-up observations in a health centre	5	CO1
2	Child Health		17	
	2.1	Child Health; Growth and Development Milestones	4	CO2
	2.2	Immunization Schedule and Nutritional Programs	4	CO2
	2.3	Management of Common Childhood Illnesses (IMNCI)	4	CO2
	2.4	Practical: Growth chart plotting and assessment	5	CO2
3	Adolescent Health		20	
	3.1	Adolescent Health Issues: Nutrition, Mental Health, Sexual Health	4	CO3
	3.2	School Health Programs	3	CO3
	3.3	Life Skills Education for Adolescents	3	CO3

	3.4	Practical: Conducting awareness session for adolescents in a school	10	CO3
4	Geriatric Health and Elder Care		21	
	4.1	Geriatric Health: Ageing Process and Common Health Problems	4	CO4
	4.2	Health Services for Elderly in India	3	CO4
	4.3	Social and Psychological Aspects of Aging	4	CO4
	4.4	Practical: Field visit to old age home and interview	10	CO4
5	Teacher Specific Content			

	Classroom Procedure (Mode of transaction) Direct Instructions: • Lecture • E-learning Interactive Instructions: • Group Assignment • Library Work and Group Discussion • Practical
Assessment Types	MODE OF ASSESSMENT
	A. Continuous Comprehensive Assessment (CCA) Theory-25 Marks • Internal Test • Assignment/Oral presentation • Quiz • In-class discussion and involvement Practical-15 Marks • Internal Test • Record • Survey report

	D. End Semester Evaluation (ESE) Theory -50 Marks Duration of Examination 1.30 hr Pattern of examination for Theory Non-MCQ • Section A - Objective type - 10/10 (10x1=10 marks) • Section B - Short Answer - 10/12 (10x2=20 marks) • Section C - Short Essay – 5/7 (5x4=20 marks) Practical-35Marks • Presentation -10 marks • Viva- 10 marks • Field visit report/Case study-10 marks • Record-5marks Duration of Examination -NA
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REFERENCES .

1. **Park, K. (2023).** *Preventive and Social Medicine* (27th ed.). Jabalpur: Banarsidas Bhanot Publishers.
2. **World Health Organization. (2022).** *Pregnancy, childbirth, postpartum and newborn care: A guide for essential practice* (3rd ed.).
3. **Ministry of Health and Family Welfare (MoHFW), Government of India. (2021).** *Reproductive, Maternal, Newborn, Child and Adolescent Health (RMNCH+A) Strategy*. New Delhi:
4. **UNICEF. (2018).** *Care for Child Development: Improving the Care of Young Children*. New York: UNICEF.
5. **Kishore, J. (2022).** *National Health Programs of India: National Policies and Legislations related to Health* (13th ed.). Century Publications.
6. **Swaminathan, M. (2012).** *Advanced Textbook on Food and Nutrition* (Vols. 1 & 2). Bangalore: The Bangalore Printing & Publishing Co.

Syllabus

SUGGESTED READINGS

1. **Srilakshmi, B. (2017).** *Nutrition Science* (4th ed.). New Delhi: New Age International Publishers.
2. **Ghai, O. P., Paul, V. K., & Bagga, A. (2019).** *Ghai Essential Pediatrics* (9th ed.). New Delhi: CBS Publishers & Distributors.
3. **Murray, R. K., et al. (2018).** *Harper's Illustrated Biochemistry* (31st ed.). New York: McGraw-Hill Education.
4. **World Health Organization. (2014).** *Health for the World's Adolescents: A Second Chance in the Second Decade*.
5. **National Institute of Social Defence (NISD). (2018).** *Manual on Geriatric Care*. New Delhi: Ministry of Social Justice and Empowerment, Government of India.



Mahatma Gandhi University

Kottayam

Faculty/ Discipline	Science				
Programme	Public Health				
Course Name	Health Systems, Policy, and Ethics				
Type of Course	MPC				
Course Code	MG5MPCPHS300				
Course Level	300				
Course Summary	This course teaches the principles of sports nutrition and hydration, focusing on macro-nutrient and micro-nutrient needs, meal planning, supplementation, and fluid strategies to support athletic performance, recovery, and health.				
Semester	5	Credits		4	Total Hours
Course Details	Learning Approach	Lecture	Practical	OJT	
		4	0	0	
Pre-requisites, if any	Nil				

COURSE OUTCOMES (CO)

CO No.	Expected Course Outcome	Learning Domains	PO No
1	Describe the components, functions, and global models of health systems, with special emphasis on the evolution of the Indian healthcare system.	U	1,2,3,4,6,7,10
2	Analyze the structure and implementation of key national health programs and compare public, private, and NGO roles in service delivery across urban and rural areas.	An, U	1,2,3,4,5,6,7,8,9,10
3	Explain the process of health policy development in India, including policy frameworks, stakeholders, and recent reforms.	U, A	1,2,3,4,5,6,7,8,9,10
4	Evaluate ethical principles and legal frameworks guiding medical and public health practice, with a focus on real-world applications.	E, U	1,2,3,4,6,7,8,9,10
Remember (K), Understand (U), Apply (A), Analyse (An), Evaluate (E), Create (C), Skill (S) Interest (I) and Appreciation (Ap).			

MGU-B.VOC. (HONOURS) CO-PO ARTICULATION MATRIX

Syllabus

CO/PO	PO 1	PO 2	PO 3	PO 4	PO 5	PO 6	PO 7	PO 8	PO 9	PO 10
CO 1	1	2	2	1	0	2	1	0	0	2
CO 2	2	2	2	1	1	3	2	1	1	2
CO 3	1	2	2	2	1	2	1	1	2	3
CO 4	1	2	1	1	0	2	2	3	1	2

'0' is No Correlation, '1' is Slight Correlation (Low level), '2' is Moderate Correlation (Medium level) and '3' is Substantial Correlation (High level).

COURSE CONTENT

Content for classroom transaction(Units)

MODULE	UNITS	COURSE DESCRIPTION	Hrs.	CO.NO.
1	Structure and Evolution of Health Systems		15	
	1.1	Overview of Health Systems: Components and Functions	5	CO1
	1.2	Evolution of Health Systems in India	5	CO1
	1.3	Health Care Models: Beveridge, Bismarck, and Market Models	5	CO1
2	National Health Programs and Service Delivery		15	
	2.1	National Health Programs: NRHM, NHM, Ayushman Bharat	6	CO2
	2.2	Role of Public, Private, and NGO sectors	5	CO2
	2.3	Urban vs. Rural Health Infrastructure	4	CO2
3	Health Policy Development and Reforms		15	
	3.1	Health Policy Framework in India	5	CO3
	3.2	Policy Formulation Process and Implementation	5	CO3
	3.3	Current Policies and Reforms (e.g., National Digital Health Mission)	5	CO3
4	Health Ethics and Legal Principles		15	
	4.1	Introduction to Health Ethics and Legal Framework	5	CO4
	4.2	Principles of Medical Ethics: Autonomy, Justice, Beneficence, Non-maleficence	5	CO4
	4.3	Ethics in Public Health Practice and Research	5	CO4
5	Teacher Specific Content			

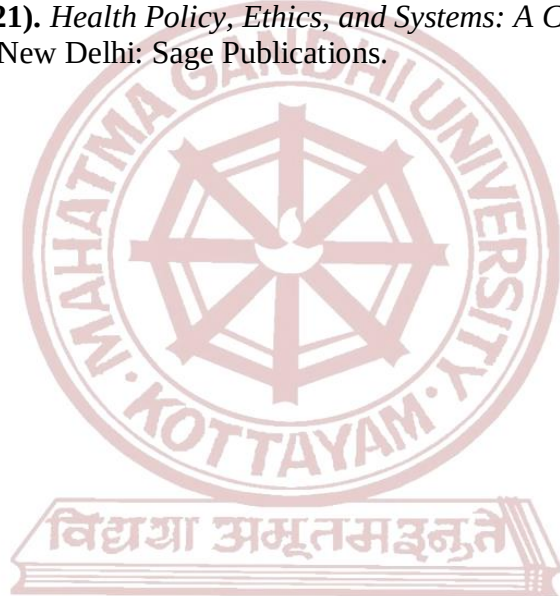
Teaching and Learning Approach	Classroom Procedure (Mode of transaction) • Direct Instructions: • Lecture • E Learning Interactive Instructions • Group Assignment • Library Work and Discussion • Practical
Assessment Types	MODE OF ASSESSMENT
	A. Continuous Comprehensive Assessment (CCA) Theory - 30 marks • Internal Test • Assignment/Oral Presentation • Quiz • In class discussion and involvement
	A. End Semester Evaluation (ESE) Theory -70 Marks Duration of Examination 2 hrs Pattern of examination for Theory Non-MCQ Section A - Short Answer – 10/12 (10x2=20 marks) Section B - Short Essay - 6/9 (6x5=30 marks) Section C - Essay – 2/4 (2x10=20 marks)

REFERENCES

1. **Park, K. (2023).** *Park's Textbook of Preventive and Social Medicine* (27th ed.). Jabalpur: Banarsidas Bhanot.
2. **WHO. (2007).** *Everybody's Business: Strengthening Health Systems to Improve Health Outcomes*. Geneva: World Health Organization.
3. **Bhatia, R. (2020).** *Essentials of Public Health: A Practical Guide*. New Delhi: CBS Publishers.
4. Ministry of Health and Family Welfare, Government of India. (2017). *National Health Policy 2017*. New Delhi.
5. **Beauchamp, T. L., & Childress, J. F. (2019).** *Principles of Biomedical Ethics* (8th ed.). New York: Oxford University Press.

SUGGESTED READING

1. **Rao, K. D., Peters, D. H., & Bandeen-Roche, K. (2006).** Towards patient-centered health services in India—A scale to measure patient perceptions of quality. *International Journal for Quality in Health Care*, 18(6), 414–421.
2. **Mills, A. (2014).** *Health Systems in Low- and Middle-Income Countries: An Economic and Policy Perspective*. Oxford University Press.
3. **Sheikh, K., George, A., & Gilson, L. (2014).** People-centred science: Strengthening the practice of health policy and systems research. *Health Research Policy and Systems*, 12(1), 19.
4. **World Health Organization. (2015).** *Health Systems Strengthening Glossary*.
5. **Sharma, R. (2021).** *Health Policy, Ethics, and Systems: A Casebook on Public Health in India*. New Delhi: Sage Publications.



MGU-B.VOC. (HONOURS)

Syllabus



Mahatma Gandhi University

Kottayam

Faculty/ Discipline	Science				
Programme	Public Health				
Course Name	Global Health and Public Health Informatics				
Type of Course	MPC				
Course Code	MG6MPCPHS300				
Course Level	300				
Course Summary	This course provides comprehensive knowledge of the causes, classification, acute care, rehabilitation, and prevention of sports injuries, enabling students to manage injuries effectively and support safe return-to-sport through evidence-based physiotherapy techniques and clinical decision-making.				
Semester	6	Credits		4	Total Hours
Course Details	Learning Approach	Lecture	Practical	OJT	
		4	0	0	
Pre-requisites, if any	Nil				

COURSE OUTCOMES (CO)

CO No.	Expected Course Outcome	Learning Domains	PO No
1	Describe the scope, indicators, and global burden of disease, focusing on major communicable and non-communicable health issues.	U	1,2,3,4,6,7,10
2	Analyze the contributions of international organizations and global health partnerships in addressing population health challenges.	An , U	1,2,3,4,5,6,7,8,9,10
3	Explain the foundations and functions of public health informatics and assess how information systems support disease surveillance and decision-making.	U, E	1,2,3,4,6,7,9,10
4	Evaluate the impact of digital innovations such as EHRs, mHealth, and AI on public health planning and service delivery.	E , A	1,2,3,4,5,6,7,8,9,10
*Remember(K), Understand(U), Apply(A), Analyse(An), Evaluate(E), Create(C), Skill(S), Interest(I) and Appreciation (Ap)			

CO-PO ARTICULATION MATRIX

CO/PO	PO 1	PO 2	PO 3	PO 4	PO 5	PO 6	PO 7	PO 8	PO 9	PO 10
CO 1	1	2	2	1	0	2	1	0	0	2
CO 2	2	2	2	2	1	3	2	1	2	2
CO 3	1	3	2	2	0	2	1	0	2	3
CO 4	2	3	2	2	1	2	1	1	2	3

‘0’ is No Correlation, ‘1’ is Slight Correlation (Low level), ‘2’ is Moderate Correlation (Medium level) and ‘3’ is Substantial Correlation (High level).

COURSE CONTENT

Content for Classroom transaction (Units)

Module	Units	Course description	Hrs	CO No.
1	Foundations of Global Health		15	
	1.1	Concept and Scope of Global Health	5	CO1
	1.2	Global Health Indicators and Burden of Disease	5	CO1
	1.3	Key Global Health Issues: Communicable and Non-Communicable Diseases	5	CO1
2	Global Partnerships and Initiatives		15	
	2.1	Role of International Organizations: WHO, UNICEF, UNDP, World Bank	5	CO2
	2.2	Global Health Initiatives: GAVI, GFATM, SDGs in Health	5	CO2
	2.3	Case Studies of Successful Global Health Campaigns	5	CO2
3	Public Health Informatics		15	
	3.1	Introduction to Public Health Informatics	5	CO3
	3.2	Health Information Systems: Components and Applications	5	CO3
	3.3	Use of Technology in Surveillance and Monitoring	5	CO3
4	Digital Innovations in Public Health		15	
	4.1	Electronic Health Records and Data Privacy	5	CO4

	4.2	mHealth and Telemedicine in Public Health	5	CO4
	4.3	Role of Big Data and Artificial Intelligence in Health Planning	5	CO4
5	Teacher Specific Content			

Teaching and Learning Approach	Classroom Procedure (Mode of transaction) Direct Instructions: • Lecture • E-learning Interactive Instructions: • Group Assignment • Library Work and Group Discussion
Assessment Types	MODE OF ASSESSMENT
	A.Continuous Comprehensive Assessment (CCA) Theory-30 Marks • Internal Test • Assignment/Oral presentation • Quiz • In-class discussion and involvement
	B. End Semester Evaluation (ESE) Theory -70 Marks Duration of Examination 2 hrs Pattern of examination for Theory Non-MCQ • Section A - Short Answer – 10/12 (10x2=20 marks) • Section B - Short Essay - 6/9 (6x5=30 marks) • Section C - Essay – 2/4 (2x10=20 marks)

REFERENCES

1. **Park, K. (2023).** *Park's Textbook of Preventive and Social Medicine* (27th ed.).
2. **Merson, M. H., Black, R. E., & Mills, A. J. (Eds.). (2012).** *Global Health: Diseases, Programs, Systems, and Policies* (3rd ed.). Burlington, MA: Jones & Bartlett Learning.
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4. **Magnuson, J. A., & Dixon, B. E. (Eds.). (2020).** *Public Health Informatics and Information Systems* (3rd ed.). Springer.
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3. **Yasnoff, W. A., O'Carroll, P. W., Koo, D., Linkins, R. W., & Kilbourne, E. (2000).** Public health informatics: Improving and transforming public health in the information age. *Journal of Public Health Management and Practice*, 6(6), 67–75.
4. **Bloland, P., & Simone, P. (2009).** Public health informatics and the potential for transforming national surveillance systems. *MMWR Supplements*, 58(1), 5–9.
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Syllabus